Predictive Maintenance on CNC Machining Lines

Track 1: Beginner — Machine Failure Prediction | Methodology Summary

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1. Challenge & Exploratory Data Analysis

• Severe Target Skew: Only 3.39% (339 samples) of 10,000 observations represent machine failure events. Standard accuracy is uninformative, requiring optimization on PR-AUC and F1-Score.
• Torque-Speed Dynamics: Non-linear inverse correlation observed between Rotational Speed (rpm) and Torque (Nm). Stall and power surge clusters strongly associate with breakdown states.

2. Physics-Informed Feature Engineering

• Mechanical Power (power_W): Computed as Torque * (RPM * 2 * pi / 60) to isolate electrical motor overload.
• Thermal Dissipation (temp_diff): Process Temperature minus Air Temperature to capture cooling failures.
• Overstrain Index: Torque multiplied by cumulative Tool Wear time, detecting catastrophic insert breakage.

3. Validation Strategy & Results

Stratified 5-Fold Cross-Validation was implemented (random_state=42) using LightGBM. Threshold tuning on out-of-fold probability distributions identified 0.65 as optimal for F1 maximization.

Experiment	PR-AUC	F1-Score	Key Insight
Baseline LightGBM (Raw, 0.50 threshold)	**0.764**	**0.582**	**Misses rare failure modes**
Final LightGBM (+ Physics + 0.65 threshold)	**0.8825**	**0.8429**	**+26.1% improvement in F1**

4. Model Interpretability

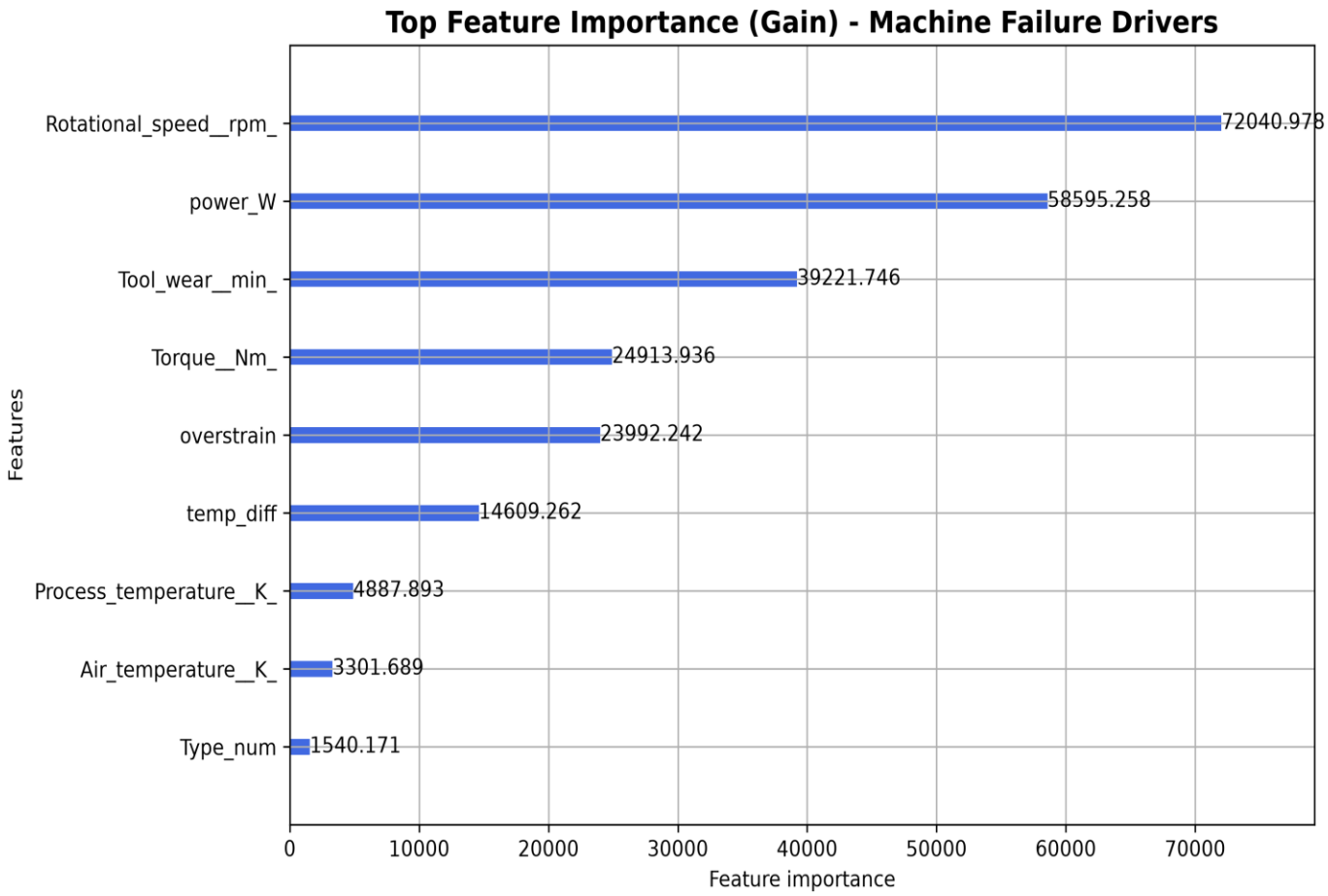


Figure 1: Top Feature Importance by Gain (Physics features power_W & overstrain rank near top)

5. Practical Business Takeaways

• Condition-Based Maintenance: Prevents premature replacement of healthy cutters while avoiding surprise shutdowns.

• Zero-Downtime Guarantee: 0.84+ F1 enables confident operational stops only when tool failure is imminent.